

MY CLASS RECORDING PLAYLIST:

<https://youtube.com/playlist?list=PLgU-6-TtNPAzpauVSPWpKFbmu5Rbc2dEQ&si=PAzhNHHluGKTAaTf>

[dijkstra class is missing, giving some supplementary links for that]
For dijkstra algo: [Dijkstra's Shortest Path Algorithm | Graph Theory](#)
[you can ignore the optimization parts and just focus on the simulation for now]

*****GRAPH*****

Topic:GRAPH: [6. Graph \[HFN\] Supplementary videos](#) [DONT SEE THE THINGS IN THIS FILE THAT ARE NOT MENTIONED IN THIS LIST]

1. Basic/RECAP:

BFS + DFS :

Lecture 1 [BFS needed, Start from 1h 17 minutes mark]:

<https://youtu.be/-kP2g7IfZr4?si=Yfi4Z7h-81PgMTxE>

Lecture 2 [DFS]:

<https://youtu.be/OlmII8stMZc?si=gYlyvZnt14-nHFsD>

*****PATH FINDING/ SHORTEEST PATH ALGO*****

2. FOR DIJKSTRA SIMULATION FOLLOW THIS VIDEO:

Single source shortest path algorithms: [USE THESE VIDEOS]

[Dijkstra's Shortest Path Algorithm | Graph Theory](#)

[Ignore the optimization parts]

[Graph Data Structure 4. Dijkstra's Shortest Path Algorithm](#)

Negative weight cycle issue: [Why Dijkstra's Algorithm Doesn't Work with Negative Weights](#)

To handle that, we use: [Must study [Bellman Ford Algorithm](#) along with dijkstra above]

[Bellman Ford Algorithm | Shortest path & Negative cycles | Graph Theory](#)

3. *****MINIMUM SPANNING TREE*****

Minimum Spanning Tree: [6.4 Dijkstra+ MST.pdf](#) [SLIDE]

[refer to my recordings Lecture [Video 5](#) , [Video 6](#) There are detailed discussion about a data structure called Disjoint Set Union, hold it for now I will discuss it in class]

1. Prims Algo: [Prim's Minimum Spanning Tree Algorithm | Graph Theory](#)
2. Kruskal Algo: [Union Find Kruskal's Algorithm](#)
3. Prims vs kruskal: <https://www.baeldung.com/cs/kruskals-vs-prim-algorithm>

4. *****GREEDY ALGO*****

1. Huffman coding[Greedy Algo]: [Follow my class: <https://youtu.be/GP8TdRC4ZKE?si=IrDptChT7ipIyiVj>]
[7. Huffman Coding](#) [all the links are given in this doc]
2. Fractional Knapsack [not dynamic, Greedy approach] [Knapsack Problem | Greedy Method | Data structure & Algorithm | Bangla Tutorial](#)

5. *****DYNAMIC PROGRAMMING*****

Dynamic Programming: [refer to my recordings Lecture 7,8

Lecture 7: https://youtu.be/2UZ_x-Nj12s?si=Bsiq2dCeAdnDPSgU

Lecture 8: <https://youtu.be/tF5djZfClLU?si=pI0prnBRFvN9WDh>

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Supplementary videos:

01 knapsack:

[0/1 Knapsack Problem Dynamic Programming](#)

LCS: [4.9 Longest Common Subsequence \(LCS\) - Recursion and Dynamic Programming](#)

ALL THE RELEVANT BUX SLIDES ARE GIVEN ALONG WITH THIS DOC IN THIS LINK:

https://drive.google.com/drive/folders/1EuJTeoAiPXXtdCI5O3k48XDK5_z4O7xU?usp=drive_link